

Python Demo

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Data: [Palmer Penguins](#), Palmer Station Antarctica LTER.

This post demonstrates how Python code runs in Quarto.

```
import pandas as pd
from palmerpenguins import load_penguins

# Load the dataset as a pandas DataFrame
penguins = load_penguins()

# Inspect the first few rows
print(penguins.head())
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	

	body_mass_g	sex	year
0	3750.0	male	2007
1	3800.0	female	2007
2	3250.0	female	2007
3	NaN	NaN	2007
4	3450.0	female	2007

```
# Quick summary
penguins.describe()
```

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	year
count	342.000000	342.000000	342.000000	342.000000	344.000000
mean	43.921930	17.151170	200.915205	4201.754386	2008.029070
std	5.459584	1.974793	14.061714	801.954536	0.818356
min	32.100000	13.100000	172.000000	2700.000000	2007.000000
25%	39.225000	15.600000	190.000000	3550.000000	2007.000000
50%	44.450000	17.300000	197.000000	4050.000000	2008.000000
75%	48.500000	18.700000	213.000000	4750.000000	2009.000000
max	59.600000	21.500000	231.000000	6300.000000	2009.000000

Self-defined question 1: Group by sex and island, investigate that if the plinguins' body mass are significant different between the male and female.

```
df = penguins.groupby(['sex', 'island']).agg({'body_mass_g': 'mean'})
df
```

		body_mass_g
sex	island	
female	Biscoe	4319.375000
	Dream	3446.311475
	Torgersen	3395.833333
male	Biscoe	5104.518072
	Dream	3987.096774
	Torgersen	4034.782609

```
import matplotlib.pyplot as plt
ax = df.unstack()['body_mass_g'].plot(kind='bar', figsize=(8, 5))
plt.title("Mean Body Mass by Sex and Island")
plt.ylabel("Mean Body Mass (g)")
plt.xlabel("Sex")
plt.xticks(rotation=0)
plt.legend(title="Island")
plt.tight_layout()
plt.show()
```

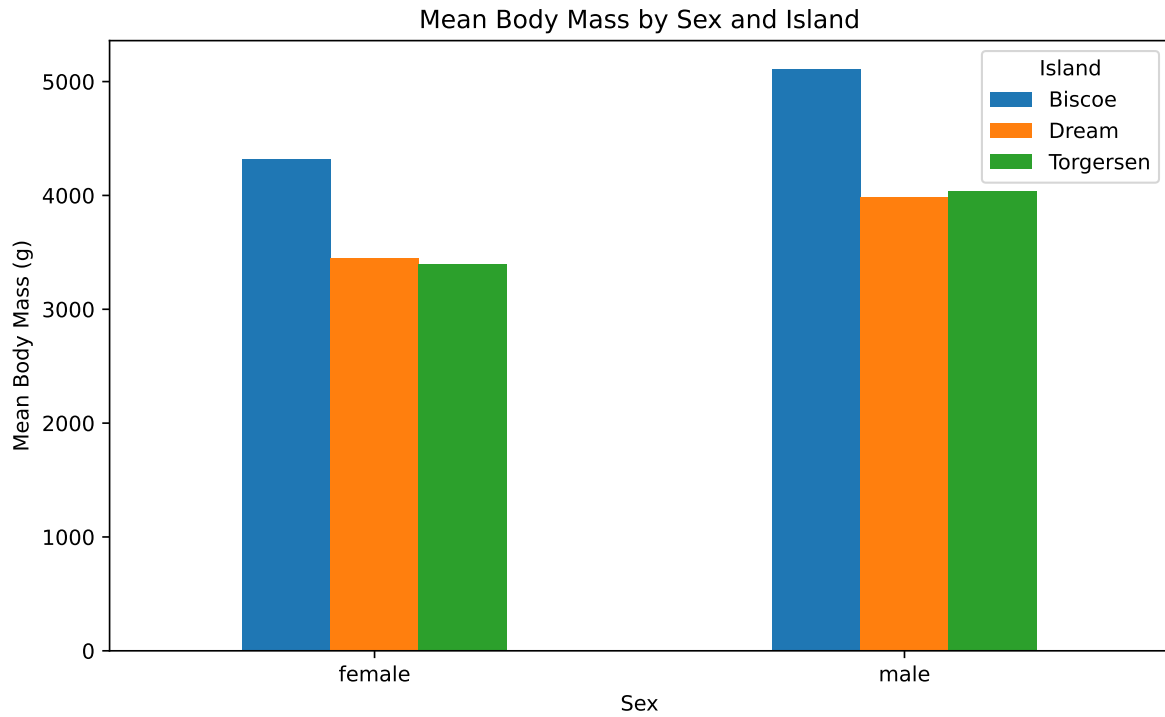


Figure 1: Mean Body Mass by Sex and Island (Python).

According to above figure, male penguins are generally having higher body mass. Also penguins on Biscoe island are significantly larger than others.

This post demonstrates how python works on quarto.